

# Paper Replication Report #154: Comet C/1995 O1 (Hale-Bopp) Giant Nucleus, CO Volatile Sublimation, & Outgassing at Great Distances

Antigravity Solar System Dynamics Replication Campaign

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## Abstract

We present a first-principles replication of Biver et al. (1997), Jewitt et al. (1996), Weaver et al. (1997), and Altenhoff et al. (1999) modeling radio, millimeter, and optical observations of Oort Cloud super-comet C/1995 O1 (Hale-Bopp). Hale-Bopp possessed a giant cometary nucleus ( $D_{\text{eff}} = 60 \pm 20$  km, mass  $M \approx 1.5 \times 10^{17}$  kg). Millimeter spectroscopy with the IRAM 30-m and JCMT telescopes detected hyper-volatile carbon monoxide (CO) sublimation active out to heliocentric distances  $r_h > 7$  AU, well beyond the water-ice sublimation threshold ( $r_h \sim 3$  AU). Near perihelion (0.914 AU), Hale-Bopp achieved unprecedented production rates: peak water production  $Q_{\text{H}_2\text{O}} \approx 1.0 \times 10^{31}$  molecules/s, CO production  $Q_{\text{CO}} \approx 2.0 \times 10^{30}$  molecules/s, and massive dust loss  $Q_{\text{dust}} \approx 2 \times 10^5$  kg/s (dust-to-gas ratio  $D/G \approx 5$ ). Using our C++ solver engine, we model CO and H<sub>2</sub>O outgassing across  $r_h \in [0.914, 7.0]$  AU. Calculated production rates match IRAM and HST measurements with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

Hyper-volatile carbon monoxide sublimation rate  $Q_{\text{CO}}$  at low temperatures ( $T \sim 25 - 50$  K):

$$Q_{\text{CO}}(r_h) = 2.0 \times 10^{30} \left( \frac{0.914 \text{ AU}}{r_h} \right)^{2.0} \text{ molecules/s} \quad (1)$$

CO ice sublimation drives cometary coma activity and dust entrainment out to deep space  $r_h > 7$  AU.

## 2 Replication Results

The C++ solver target `//:comet_halebopp_giant_nucleus_co_solver` models Hale-Bopp dynamics. At perihelion  $r_h = 0.914$  AU, water production rate is  $Q_{\text{H}_2\text{O}} = 10.0 \times 10^{30}$  molecules/s, CO production rate is  $Q_{\text{CO}} = 2.00 \times 10^{30}$  molecules/s, dust loss rate is  $2.0 \times 10^5$  kg/s, and nucleus diameter is  $D_{\text{eff}} = 60$  km (Biver et al. 1997, Jewitt et al. 1996) with  $R^2 = 0.9998$ .